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DISPLAY APPARATUS

Abstract

A display apparatus according to an embodiment of the disclosure may include a display module, a cabinet to which the display module is coupled, a circuit case coupled to the cabinet and receiving a printed circuit board, a power connector arranged on one wall of the circuit case to supply power to the printed circuit board, a cover detachably coupled to the power connector, and a locking member coupled to the one wall of the circuit case on which the power connector is arranged and configured to lock the cover by preventing rotation of the cover when the cover is coupled to the power connector.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of International Application No. PCT/KR2023/016393, filed on Oct. 20, 2023, in the Korean Intellectual Property Receiving Office, which claims priority to Korean Patent Application No. 10-2022-0183593, filed on Dec. 23, 2022, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

BACKGROUND

1. Field

[0002] The present disclosure relates to a display apparatus, and more particularly, to a display apparatus having increased reliability and ease of use by attaching a cover or wire conduit to a power connector.

2. Description of Related Art

[0003] A display device is a type of output device that converts acquired or stored electrical information into visual information and displays such information to a user.

[0004] To realize a large-sized screen, a display device may include a plurality of display modules and a cabinet supporting the plurality of display modules by continuously tiling them in the up, down, left, and right directions. The cabinets supporting the plurality of display modules may be provided as a plurality, and the plurality of cabinets may be arranged to be continuously tiled in the up, down, left, and right directions as are the plurality of display modules. The plurality of display modules supported by the plurality of cabinets may form a single large-sized screen.

[0005] In addition, the display device includes a circuit case coupled to the cabinet which may supply power to the plurality of display modules via a power supply arranged therein. One side of the circuit case may be provided with a power connector for receiving power from an external source.

[0006] The display device may be installed outdoors rather than indoors. In particular, when the power connector is exposed to the outside, the power connector may be damaged or failure due to weather conditions such as rain or snow, or animals contacting the power connector. Therefore, it is important that, when the power connector is not in use, the power connector may be prevented from damage and failure by covering the power connector with a cover that may not be easily separated from the power connector.

[0007] On the other hand, even when a wire conduit supplying power from an external source is connected to the power connector outdoors, the wire conduit need not to be easily separated from the power connector in order to reliably supply power to the display module.

SUMMARY

[0008] According to an embodiment of the disclosure, a display apparatus may include a display module, a cabinet to which the display module is coupled, a circuit case coupled to the cabinet and receiving a printed circuit board, a power connector arranged on one wall of the circuit case to supply power to the printed circuit board, a cover detachably coupled to the power connector, and a locking member coupled to the one wall of the circuit case on which the power connector is

arranged, and configured to lock the cover by preventing rotation of the cover when the cover is coupled to the power connector.

[0009] According to an embodiment of the disclosure, a display apparatus may include a display module, a cabinet to which the display module is coupled, a circuit case coupled to the cabinet and configured to receive a printed circuit board, a power connector arranged on one wall of the circuit case to supply power to the printed circuit board, a wire conduit connectable to the power connector to supply power to the display module from an outside, and a locking member coupled to the one wall of the circuit case on which the power connector is arranged, the locking member sliding toward the wire conduit to lock the wire conduit when the wire conduit is coupled to the power connector.

[0010] According to an embodiment of the disclosure, a display apparatus may include a display module configured to display image information facing forward, a cabinet coupled to a rear side of the display module, a circuit case coupled to the cabinet and configured to receive a printed circuit board, a power connector arranged on a side wall of the circuit case to supply power to the printed circuit board, a cover detachably coupled to the power connector, and a locking member coupled to the side wall of the circuit case and configured to contact the cover to lock the cover so as to prevent rotation of the cover when the cover is coupled to the power connector.

[0011] According to an aspect of the present disclosure, by locking the cover or wire conduit coupled to the power connector in two stages, the display apparatus may be provided in which damage and failure may be prevented by preventing the cover or wire conduit from being detached when not needed.

[0012] According to an aspect of the present disclosure, by locking the cover or wire conduit coupled to the power connector by a simple structure, the display apparatus may be provided in which the cover or wire conduit may be easily locked and detached as needed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following detailed description, taken in conjunction with the accompanying drawings.

[0014] FIG. 1 is a perspective view of a display apparatus according to one or more embodiments.

[0015] FIG. 2 is an exploded perspective view of the display apparatus according to one or more embodiments.

[0016] FIG. 3 is a partially enlarged view of the display apparatus of FIG. 2.

[0017] FIG. 4 is a rear exploded perspective view of the display apparatus according to one or more embodiments.

[0018] FIG. 5 is a rear exploded perspective view of a display module in the display apparatus according to one or more embodiments.

[0019] FIG. 6 is a front view of a cabinet in the display apparatus according to one or more embodiments.

[0020] FIG. 7 is a partially enlarged view of the display apparatus according to one or more embodiments.

[0021] FIGS. 8 and 9 are exploded perspective views of some configurations of the display apparatus shown in FIG. 7.

[0022] FIGS. 10 to 17 are views illustrating a process in which a locking member locks a cover in the display apparatus according to one or more embodiments.

[0023] FIG. 18 is a side cross-sectional view of the display apparatus according to one or more embodiments.

[0024] FIGS. **19** to **22** are views illustrating a process in which the locking member locks the cover in the display apparatus according to one or more embodiments.

DETAILED DESCRIPTION

[0025] Embodiments described in the disclosure and configurations shown in the drawings are merely examples of the embodiments of the disclosure and may be modified in various different ways at the time of filing of the present application to replace the embodiments and drawings of the disclosure.

[0026] In addition, the same reference numerals or signs shown in the drawings of the disclosure indicate elements or components performing substantially the same function.

[0027] Also, the terms used herein are used to describe the embodiments and are not intended to limit and/or restrict the disclosure. The singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. In this disclosure, the terms “including,” “having,” and the like are used to specify features, figures, steps, operations, elements, components, or combinations thereof, but do not preclude the presence or addition of one or more of the features, figures, steps, operations, elements, components, or combinations thereof.

[0028] It will be understood that, although the terms first, second, primary, secondary, etc., may be used herein to describe various elements, but elements are not limited by these terms. These terms are only used to distinguish one element from another element. For example, without departing from the scope of the disclosure, a first element may be termed as a second element, and a second element may be termed as a first element. The term of “and/or” includes a plurality of combinations of relevant items or any one item among a plurality of relevant items.

[0029] Further, as used in the disclosure, the terms “front,” “rear,” “top,” “bottom,” “side,” “left,” “right,” “upper,” “lower,” and the like are defined with reference to the drawings, and are not intended to limit the shape and position of each component.

[0030] Various embodiments of the present disclosure will be described in detail below with reference to the accompanying drawings.

[0031] FIG. **1** is a perspective view of a display apparatus according to one or more embodiments. FIG. **2** is an exploded perspective view of the display apparatus according to one or more embodiments. FIG. **2** is a view illustrating a cabinet and a display module separated from the cabinet, viewed from the front, in the display apparatus according to one or more embodiments.

[0032] FIG. **1** shows an X-axis, a Y-axis, and a Z-axis direction perpendicular to each other, wherein the X-axis direction may refer to a left-to-right direction, the Y-axis direction may refer to an up-and-down direction, and the Z-axis direction may refer to a front-to-back direction. However, the meanings of the X-axis, Y-axis, and Z-axis directions are not limited to the example described above and may be varied.

[0033] Referring to FIGS. **1** and **2**, a display apparatus **1** is a device that displays information, materials, data, and the like in the form of characters, shapes, graphs, images, and the like, and a billboard, a signboard, a screen, a television, a monitors, and the like may be implemented as the display apparatus **1**. The display apparatus **1** may be mounted on a wall or ceiling, or may be mounted on an indoor or outdoor ground by means of a stand (not shown).

[0034] The display apparatus **1** may include a display module **100** that displays a screen, a cabinet **200** coupled to a rear side of the display module **100** to support the display module **100**, and a circuit case **300** coupled to the cabinet **200** and accommodating electrical components and/or printed circuit boards (PCBs) therein.

[0035] In the display apparatus **1** according to one or more embodiments, six display modules **100** may be coupled to one cabinet **200**. Specifically, two display modules **100** may be coupled to one cabinet **200** in the left-to-right direction X, and three display modules **100** may be coupled in the up-and-down direction Y. In other words, the display modules **100** may be coupled to the cabinet **200** in a 2*3 matrix. However, the number of display modules **100** coupled to the cabinet **200** is not limited. Two or more display modules **100** may be coupled to one cabinet **200**.

[0036] Referring to FIG. 1, the display apparatus **1** according to one or more embodiments may include eight cabinets **200** and 48 display modules **100** coupled to the eight cabinets **200**. As such, the display apparatus **1** may realize a large-sized screen by arranging a plurality of cabinets **200**, to which a plurality of display modules **100** are coupled, adjacent to each other in the left-to-right direction X and the up-and-down direction Y. The number of the plurality of cabinets **200** arranged adjacent to each other in the left-to-right direction X and the up-and-down direction Y is not limited, and as described above, the number of the plurality of display modules **100** coupled to the cabinet **200** is also not limited.

[0037] Referring to FIG. 2, the display module **100** may be detachably coupled to the cabinet **200**. The display module **100** may be detachable from the cabinet **200** toward a front side of the cabinet **200**.

[0038] The display module **100** may be coupled to the cabinet **200** by magnetic forces. The cabinet **200** and the circuit case **300** may include a plurality of magnets **10** arranged on the front thereof. The plurality of magnets **10** may include a plurality of first magnets **10** arranged on a front surface of the cabinet **200**. The plurality of magnets **10** may include a plurality of second magnets **10** arranged on a front surface of the circuit case **300**. The display module **100** may include a reinforcing member **140** (see FIG. 5) arranged on a rear side thereof and coupled by magnetic attraction to the plurality of first magnets **10** and the plurality of second magnets **10**. The plurality of first magnets **10** arranged on the cabinet **200** and the plurality of second magnets **10** arranged on the circuit case **300** may be of the same configuration as each other. The plurality of first magnets **10** and the plurality of second magnets **10** may be collectively referred to as the plurality of magnets **10**.

[0039] The cabinet **200** may include frames **210**, **220**, **230** and **240** (see FIG. 6) forming an outer edge of the cabinet **200**. The circuit case **300** may be disposed inside the frames **210**, **220**, **230** and **240**. The circuit case **300** may include a second connector **301** corresponding to a first connector **112** (see FIG. 4) included in each of the plurality of display modules **100**. The circuit case **300** may include a first case **310** having a hexahedral shape with an open side, and a second case **320** arranged to cover the open side of the first case **310**.

[0040] The plurality of display modules **100** may be electrically connected to each other by coupling the first connector **112** of each of the plurality of display modules **100** and the second connectors **301** of the circuit case **300**. In addition, the plurality of display modules **100** may receive power from a power supply device disposed inside the circuit case **300**.

[0041] The display apparatus according to one or more embodiments may further include a locking member **500** coupled to the circuit case **300**. When a cover **600** or **720** is coupled or mounted to a power connector **360**, which will be described later, the locking member **500** may lock the cover **600** or **720** to prevent from being removed from the power connector **360** (see FIGS. 16, 17, and 22).

[0042] The display apparatus according to one or more embodiments may further include a connector cover **600**. The connector cover **600** may cover the power connector **360**, which will be described later.

[0043] FIG. 3 is a partially enlarged view of the display apparatus of FIG. 2. FIG. 3 is an enlarged view of region 'A' of FIG. 2.

[0044] Referring to FIG. 3, the display module **100** may include a substrate **110** (see FIG. 5) and a holder **120** that covers the sides and rear of the substrate **110** and supports the substrate **110**. The substrate **110** may be attached to a front surface of the holder **120** via an adhesive, double-sided adhesive tape, or the like, or may be coupled to the holder **120** via a fastening member such as a screw.

[0045] A plurality of light-emitting diodes (LEDs) **111** may be mounted on a front surface of the substrate **110**. The substrate **110** may be formed of a material, such as glass, polyimide (PI), frame retardant 4 (FR4), or the like.

[0046] Each of the plurality of LEDs **111** may include a red LED, a green LED, and a blue LED packaged into a single pixel. The plurality of LEDs **111** may be mounted on the substrate **110** in a matrix form. The plurality of LEDs **111** may be arranged to have a certain spacing, and the spacing between the plurality of LEDs **111** may be varied depending on the resolution and size of the display apparatus **1**.

[0047] The display module **100** may include a cover plate **130** configured to cover the front side of the substrate **110**. The cover plate **130** may include a plurality of LED holes **131** each corresponding to the plurality of LEDs **111** mounted on the substrate **110**. The plurality of LED holes **131** may allow the plurality of LEDs **111** to be exposed on a front surface of the cover plate **130**.

[0048] The substrate **110** may be received within the cover plate **130** and the holder **120**. The cover plate **130** and the holder **120** may be coupled together to form an internal space that receives the substrate **110**.

[0049] The cover plate **130** may include a louver portion **132**. The louver portion **132** may include a horizontal louver portion **133** and a vertical louver portion **134**. The louver portion **132** may reduce or block moisture from entering the plurality of LEDs **111**. In addition, the cover plate **130** may include a roof portion **135** arranged on an upper side of the plurality of LEDs **111**. The roof portion **135** may protrude forwardly from the cover plate **130** to prevent water from directly entering the plurality of LEDs **111** from above.

[0050] The cover plate **130** may include a fastening hole **136** into which a fastening member **30** for coupling the cover plate **130**, the substrate **110**, and the holder **120** is inserted, and a fixing hole **137** into which a fixing member **20** is inserted.

[0051] FIG. **4** is a rear exploded perspective view of the display apparatus according to one or more embodiments. FIG. **4** is a rear view illustrating the cabinet and the display module separated from the cabinet in the display apparatus according to one or more embodiments. FIG. **5** is a rear exploded perspective view of the display module in the display apparatus according to one or more embodiments.

[0052] Referring to FIGS. **4** and **5**, the display module **100** may be detachable toward the front side of the cabinet **200**. When a force greater than or equal to the magnetic attraction between the cabinet **200** and the display module **100** is applied to the display module **100** in a direction opposite to the magnetic attraction, the display module **100** may be detached from the cabinet **200**.

[0053] As described above, the display module **100** may be coupled to the cabinet **200** by magnetic forces. The plurality of magnets **10** (see FIG. **2**) may be arranged on the front surface of the cabinet **200**, and the reinforcing member **140**, which is attracted by the magnetic attraction of the plurality of magnets **10**, may be arranged on the rear side of the display module **100**.

[0054] The reinforcing member **140** may be coupled to the rear side of the holder **120**. The reinforcing member **140** may be disposed on an edge of the rear surface of the holder **120**. The reinforcing member **140** may include a first reinforcing member **141**, a second reinforcing member **142**, a third reinforcing member **143**, and a fourth reinforcing member **144**. The first reinforcing member **141** may be disposed adjacent to an upper edge of the rear surface of the holder **120** and may extend in a horizontal direction. The second reinforcing member **142** may be disposed adjacent to a lower edge of the rear surface of the holder **120** and may extend in the horizontal direction. The third reinforcing member **143** may be disposed adjacent to a left edge of the rear surface of the holder **120** and may extend in a vertical direction. The fourth reinforcing member **144** may be disposed adjacent to a right edge of the rear surface of the holder **120** and may extend in the vertical direction.

[0055] The reinforcing member **140** may increase the strength of the display module **100** by being coupled to the rear surface of the holder **120**. The reinforcing member **140** may prevent the display module **100** from warping. The reinforcing member **140** may prevent physical deformation of the display module **100**.

[0056] The reinforcing member **140** may be formed of a material that is attracted by the magnetic force of the magnets **10**. The reinforcing member **140** may be formed of a material that is magnetizable by an external magnetic force. The reinforcing member **140** may be attracted to the magnet **10** and to each other by the polarity acquired by the reinforcing member **140** from the external magnetic force. The reinforcing member **140** may be configured to include a magnetic material that interacts with the magnets **10**. For example, the reinforcing member **140** may include iron (Fe).

[0057] The display module **100** may include the fixing member **20**. The fixing member **20** may be disposed one at a rear upper side and one at a rear lower side of the display module **100**. However, the positions of the fixing member **20** are not limited to those shown.

[0058] The display module **100** may include a first connector hole **128** that exposes the first connector **112**, arranged on the rear side of the substrate **110**, to the rear side of the display module **100**. A sealing groove **125** for sealing the first connector hole **128** may be formed in the holder **120**, and a sealing member **150** may be arranged in the sealing groove **125**.

[0059] The sealing member **150** may be arranged to correspond to the shape of the sealing groove **125**. The sealing member **150** may be provided in an approximately ring shape. The sealing member **150** may be made of a material capable of volume change, and may be configured to include, for example, rubber.

[0060] According to one or more embodiments, the sealing member **150** may be attached to the rear surface of the display module **100**, but is not limited thereto. The sealing member **150** may also be attached to the front surface of the circuit case **300** so as to correspond to the sealing groove **125**.

[0061] According to one or more embodiments, the fixing member **20** may be provided in the form of a wing nut. The fixing member **20** may be insertable into a holder hole **121** of the holder **120**.

[0062] FIG. **6** is a front view of a cabinet in the display apparatus according to one or more embodiments.

[0063] Referring to FIG. **6**, the cabinet **200** may be formed by coupling four frames **210**, **220**, **230** and **240** extending in one direction. The four frames **210**, **220**, **230** and **240** may be coupled to each other by four corner brackets **250a**, **250b**, **250c** and **250d**.

[0064] The frames **210**, **220**, **230** and **240** may include a first frame **210** forming an upper edge of the cabinet **200**, a second frame **220** forming a lower edge of the cabinet **200**, a third frame **230** forming a left edge of the cabinet **200**, and a fourth frame **240** forming a right edge of the cabinet **200**. The first frame **210** and the second frame **220** may be arranged in the horizontal direction, and the third frame **230** and the fourth frame **240** may be arranged in the vertical direction.

[0065] The first frame **210** and the fourth frame **240** may be coupled by the first corner bracket **250a**. The first frame **210** and the third frame **230** may be coupled by the second corner bracket **250b**. The second frame **220** and the third frame **230** may be coupled by the third corner bracket **250c**, and the second frame **220** and the fourth frame **240** may be coupled by the fourth corner bracket **250d**.

[0066] The first frame to the fourth frame **210**, **220**, **230** and **240** may be coupled to each other by the first corner bracket to the fourth corner bracket **250a**, **250b**, **250c** and **250d** to have a rectangular frame shape.

[0067] The circuit case **300** may be located at an inner center of the frames **210**, **220**, **230** and **240**. The circuit case **300** may be coupled to the frames **210** and **220** by a first connection bracket **340** and a second connection bracket **350** arranged on upper and lower sides of the circuit case **300**, respectively.

[0068] The circuit case **300** may include the first case **310** that forms the front and sides of the circuit case **300** and is open at the rear, and the second case **320** configured to cover the open rear of the first case **310**. The first case **310** and the second case **320** may be coupled together to form the circuit case **300**. The first case **310** and the second case **320** may be coupled together such that

their interiors are hermetically sealed. The first case **310** and the second case **320** may be coupled together such that water ingress from the outside is blocked.

[0069] A second connector hole **311a** may be arranged on a front side of the first case **310**. The first case **310** may include a rib **313** arranged along a perimeter of the second connector hole **311a** and protruding forward. The second connector **301** may be disposed inside the second connector hole **311a**.

[0070] The second connector hole **311a**, the second connector **301**, and the rib **313** may be provided corresponding to the number of display modules **100** coupled to the cabinet **200**.

[0071] A power supply device for supplying power to the display module **100**, a control board for controlling the display module **100**, and the like may be provided with the circuit case **300**. In the present disclosure, the electrical components may include the power supply device, the control board, and various electronic components described above.

[0072] The circuit case **300** may include a support bracket **330** configured to support the display module **100**. The support bracket **330** may be coupled to opposite sides of the circuit case **300**. Furthermore, the support bracket **330** may be provided on each of the upper and lower sides on opposite sides of the circuit case **330**. According to one or more embodiments, four support brackets **330** may be coupled to the circuit case **330**.

[0073] The cabinet **200** may include the plurality of magnets **10**. In addition, the circuit case **300** may include the plurality of magnets **10**.

[0074] For example, the cabinet **200** may include the plurality of magnets **10** arranged to be spaced apart from each other on the front surfaces of the first frame to the fourth frame **210**, **220**, **230** and **240**.

[0075] In addition, the circuit case **300** may include the plurality of magnets **10** arranged to be spaced apart from each other on the front side of the circuit case **300**.

[0076] The cabinet **200** and the circuit case **300** according to one or more embodiments may include 10 magnets **10** per one display module **100**. The cabinet **200** and the circuit case **300** may include a total of 60 magnets **10**. The cabinet **200** may include 38 magnets **10**, and the circuit case **300** may include 22 magnets **10**. However, the number and location of the magnets **10** are not limited, and the number and location of the magnets **10** may be varied.

[0077] For example, to support the display module **100** coupled to the upper right side of the cabinet **200**, the cabinet **200** may include first to seventh magnets **10a**, **10b**, **10c**, **10d**, **10e**, **10f**, and **10g**, and the circuit case **300** may include eighth to tenth magnets **10h**, **10i** and **10j**.

[0078] The plurality of magnets **10**, as described above, may couple the display module **100** to the cabinet **200** by magnetically attracting the reinforcing member **140**, which is made of a magnetizable material and disposed on the rear surface of the display module **100**.

[0079] The display apparatus according to one or more embodiments may further include the locking member **500** coupled to the circuit case **300**. When the cover **600** or **720** is coupled or mounted to the power connector **360**, which will be described later, the locking member **500** may lock the cover **600** or **720** such that the cover **600** or **720** is not detached from the power connector **360** (see FIGS. **16**, **17**, and **22**).

[0080] Although the locking member **500** is shown in the drawings as being coupled to a lower portion of a side wall **311** of the first case **310**, the position of the locking member **500** may be varied depending on the position of the power connector **360**. Thus, the locking member **500** may also be positioned on an upper portion of the side wall **311**, a rear wall **321**, or the like of the circuit case **300**. However, for ease of description, the present disclosure illustrates that the power connector **360** is arranged on the side wall **311**, and the locking member **500** is also disposed on the side wall **311**.

[0081] The display apparatus according to one or more embodiments may further include the locking member **500** coupled to the circuit case **300**. When the cover **600** or **720** is coupled or mounted to the power connector **360**, which will be described later, the locking member **500** may

lock the cover **600** or **720** to prevent the cover **600** or **720** from being detached from the power connector **360** (see FIGS. **16**, **17**, and **22**).

[0082] FIG. **7** is a partially enlarged view of the display apparatus according to one or more embodiments.

[0083] Referring to FIG. **7**, the display apparatus according to one or more embodiments of the present disclosure may include the cover **600**. The cover **600** may cover the power connector **360** arranged on one wall, such as the side wall **311** of the circuit case **300**. For example, when the power connector **360** is attached to the side wall **311** of the circuit case **300**, the cover **600** may cover the power connector **360** from the side wall **311** side of the circuit case **300**.

[0084] By covering the power connector **360**, the cover **600** may prevent water and/or foreign substances from entering the power connector **360** even when the display apparatus is installed outdoors, and may prevent wild animals from damaging the power connector **360**. The cover **600** may be detachably coupled to the power connector **360**. The cover **600** may be referred to as a cap cover **600**.

[0085] The display apparatus according to one or more embodiments may include the locking member **500**. The locking member **500** may lock the cap cover **600** covering the power connector **360** on the one wall **311** side of the circuit case **300**. The locking member **500** may be coupled to the one wall **311** of the circuit case **300** in which the power connector **360** is provided, and when the cap cover **600** is coupled to the power connector **360**, and may slide toward the cap cover **600** to lock the cap cover **600**. For example, when the power connector **360** is attached to the side wall **311** of the circuit case **300**, the locking member **500** may lock the cap cover **600** on the side wall **311** side of the circuit case **300**. The locking member **500** may contact the cap cover **600** to prevent the cap cover **600** from rotating.

[0086] FIGS. **8** and **9** are exploded perspective views of some configurations of the display apparatus shown in FIG. **7**.

[0087] Referring to FIGS. **8** and **9**, the display apparatus according to one or more embodiments may include the power connector **360** attached to the one wall **311** of the circuit case **300**. For example, the power connector **360** may be disposed on the side wall **311** of the circuit case **300**. The power connector **360** may also be provided in a plurality. For example, the power connectors **360** may be provided on each of both side walls **311** of the circuit case **300**. However, the arrangement of the power connector **360** is not limited to the examples described above, and may also be arranged on the rear wall **321** or a lower wall **312**.

[0088] The power connector **360** may include a connecting portion **361** and an outer wall **362**. The connecting portion **361** may be electrically connected to a wire conduit **700** (or an electrical wire conduit). The connecting portion **361** may include a connecting groove **361a** for connection with the wire conduit **700**. However, the present disclosure is not limited thereto, and a connecting protrusion may be formed in the connecting portion **361**. The outer wall **362** may surround the connecting portion **361**. The outer wall **362** may protect the connecting portion **361**.

[0089] A mounting recess **363** may be formed on the outer wall **362** of the power connector **360**. The mounting recess **363** may be formed such that a mounting protrusion **630** formed on the cap cover **600** may be mountable when the cap cover **600** is primarily locked to the power connector **360**. For example, the mounting recess **363** may be formed on an upper and/or lower portion of the outer wall **362**. The location, number, and shape of the mounting recess **363** are not limited to the examples described above.

[0090] The display apparatus according to one or more embodiments may include a first fixing member **380**. The first fixing member **380** may allow the power connector **360** to be held to the one wall **311** of the circuit case **300**. For example, the power connector **360** may be fitted into a fixing hole **380a** of the first fixing member **380**. The outer wall **362** of the power connector **360** may be fitted into the fixing hole **380a**. The first fixing member **380** may function as a nut.

[0091] The display apparatus according to one or more embodiments may include the locking

member **500**. When the cap cover **600** or the wire conduit **700** is coupled to the power connector **360**, the locking member **500** may lock the cap cover **600** or the wire conduit **700** to prevent the cap cover **600** or the wire conduit **700** from being rotated and detached.

[0092] The locking member **500** may include a fastening portion **510**. The fastening portion **510** may be coupled to the one wall **311** of the circuit case **300** where the power connector **360** is positioned. The fastening portion **510** may be fastened to the one wall **311** of the circuit case **300** via a second fixing member **370** and a screw **560**. The fastening portion **510** may be provided with a fastening hole **511**. For example, the second fixing member **370** may penetrate the fastening hole **511**, and the second fixing member **370** may be inserted into a fixing hole **311b** formed in the side wall **311** of the circuit case **300** to attach the locking member **500** to the circuit case **300**. The fastening hole **511** may extend in one direction. For example, the fastening hole **511** may extend in the up-and-down direction.

[0093] The fastening portion **510** may allow the locking member **500** to slide. For example, the locking member **500** may slide along the fastening hole **511** formed in the fastening portion **510**. For example, the fastening portion **510** may slide in the up-and-down direction. The sliding direction of the fastening portion **510** is not limited to the example described above. The fastening portion **510** may be a sliding portion. The fastening hole **511** may be a sliding hole.

[0094] The locking member **500** may include an inclined portion **520**. The inclined portion **520** may extend from the fastening portion **510**. The inclined portion **520** may be inclined in a direction away from the side wall **311** of the circuit case **300** from the fastening portion **510**. For example, the inclined portion **520** may be formed to be inclined downwardly from the fastening portion **510**. The inclined portion **520** may extend to connect with the locking portion **530**. One end of the inclined portion **520** may be connected to the fastening portion **510**, and the other end may be connected to the locking portion **530**.

[0095] The locking member **500** may include the locking portion **530**. The locking portion **530** may extend from the inclined portion **520** to lock the cap cover **600** or the wire conduit **700**. The locking portion **530** may be contactable with the cap cover **600** or the wire conduit **700** to prevent rotation of the cap cover **600** or the wire conduit **700**. For example, the locking portion **530** may contact a retaining protrusion **620** of the cap cover **600** or a retaining protrusion **720** of the wire conduit **700** to prevent the cap cover **600** or the wire conduit **700** from rotating. The locking portion **530** may be connected to the inclined portion **520** on a side far from the side wall **311**.

[0096] A bent portion **540** may include a contact portion **541**. The contact portion **541** may be bent and extend from the locking portion **530**. When the locking member **500** is coupled to the side wall **311**, the contact portion **541** may extend from the locking portion **530** toward the side wall **311** of the circuit case **300**. The contact portion **541** may be spaced apart from the side wall **311** of the circuit case **300** by a given distance to prevent the side wall **311** of the circuit case **300** from being scratched when the locking member **500** slides.

[0097] The contact portion **541** may extend from a lower end of the locking portion **530** toward the side wall **311**. The contact portion **541** may stably prevent the cap cover **600** or the wire conduit **700** from rotating by making surface contact with the retaining protrusion **620** of the cap cover **600** or the retaining protrusion **720** of the wire conduit **700**.

[0098] The bent portion **540** may include a reinforcing portion **542**. The reinforcing portion **542** may be bent and extend from the locking portion **530**. When the locking member **500** is coupled to the side wall **311**, the reinforcing portion **542** may extend from the locking portion **530** toward the side wall **311** of the circuit case **300**. The reinforcing portion **542** may be spaced apart from the side wall **311** of the circuit case **300** by a given distance to prevent the side wall **311** of the circuit case **300** from being scratched when the locking member **500** slides.

[0099] The reinforcing portion **542** may extend from both side ends of the locking portion **530** toward the side wall **311**. The reinforcing portion **542** may add the strength of the locking member **500**. For example, the reinforcing portion **542** may increase the strength of the contact portion **541**

that contacts the retaining protrusion **620** of the cap cover **600** or the retaining protrusion **720** of the wire conduit **700**. The reinforcing portion **542** may also function as a handle when the locking member **500** is to be moved by a user.

[0100] One of the contact portion **541** and the reinforcing portion **542** included in the bent portion **540** may be eliminated.

[0101] The locking member **500** may further include a handle **550**. The handle **550** of the locking member **500** may be arranged at one end of the locking member **500**. For example, the handle **550** of the locking member **500** may be formed on an upper end of the locking member **500**. The handle **550** of the locking member **500** may be connected to the fastening portion **510**. The user may utilize the handle **550** of the locking member **500** to slide the locking member **500**. The handle **550** of the locking member **500** may extend from the fastening portion **510** in a direction away from the side wall **311**.

[0102] The display apparatus according to one or more embodiments may further include the second fixing member **370**. The second fixing member **370** may allow the locking member **500** to be coupled to the one wall **311** of the circuit case **300**. For example, the second fixing member **370** may allow the locking member **500** to be fastened to the side wall **311** of the circuit case **300**. The second fixing member **370** may be inserted into the fixing hole **311b** formed in the side wall **311**. The screw **560** may be inserted into a fixing portion **371** of the second fixing member **370** (see FIG. 18).

[0103] The display apparatus according to one or more embodiments may further include the screw **560**. The screw **560** may allow the locking member **500** to be coupled to the one wall **311** of the circuit case **300**. For example, the screw **560** may allow the locking member **500** to be fastened to the side wall **311** of the circuit case **300**. The screw **560** may be inserted into the fastening hole **511** formed in the locking member **500** and the fixing portion **371** formed in the second fixing member **370** to allow the locking member **500** to be secured to the side wall **311** of the circuit case **300**. The locking member **500** may be slidably moved in a state of being secured to the side wall **311** by the screw **560**.

[0104] The display apparatus according to one or more embodiments of the present disclosure may include the cap cover **600**. The cap cover **600** may cover the power connector **360** when the wire conduit **700** is not coupled to the power connector **360**. The cap cover **600** may be releasably coupled to the power connector **360**.

[0105] The cap cover **600** may include a cover portion **610**. The cover portion **610** may form an exterior of the cap cover **600** such that the power connector **360** is received within the cap cover **600**. A receiving portion **600a** may be formed inside the cap cover **600** to receive the power connector **360**. A shape of the cover portion **610** may correspond to a shape of the power connector **360**.

[0106] The cover portion **610** may include a first cover portion **611** and a second cover portion **612**. When the cap cover **600** is coupled to the power connector **360**, the first cover portion **611** may be disposed in line with the side wall **311** of the circuit case **300**. The first cover portion **611** may cover the connecting portion **361**. The second cover portion **612** may extend from a perimeter of the first cover portion **611** toward the side wall **311**. The second cover portion **612** may be formed corresponding to the outer wall **362** of the power connector **360**. The second cover portion **612** may have an inner surface **612b** and an outer surface **612a**.

[0107] The cap cover **600** may include the mounting protrusion **630**. The mounting protrusion **630** may protrude from the inner surface **612b** of the cover portion **610** and be mounted to the power connector **360**. For example, the mounting protrusion **630** may protrude from the inner surface **612b** of the second cover portion **612**. When the cap cover **600** is primarily locked to the power connector **360**, the mounting protrusion **630** may be mounted in the mounting recess **363**. The mounting protrusions **630** may be provided in a plurality.

[0108] The cap cover **600** may include the retaining protrusion **620**. The retaining protrusion **620**

may protrude from the outer surface **612a** of the cover portion **610** and be retained and secondarily locked by the locking member **500**. For example, the retaining protrusion **620** protrudes from the outer surface **612a** of the second cover portion **612**. After the cap cover **600** is primarily locked to the power connector **360**, the cap cover **600** may be secondarily locked as the locking member **500** slides to contact the mounting protrusion **630** and the locking portion **530**. The retaining protrusions **620** may be provided in a plurality. The retaining protrusion **620** may protrude from the outer surface **612a** of the second cover portion **612** on both sides.

[0109] The cap cover **600** may include a handle **640**. The handle **640** of the cap cover **600** may be formed on the first cover portion **611**. The handle **640** of the cap cover **600** may protrude from the first cover portion **611** in a direction away from the side wall **311** to allow the user to easily move the cap cover **600**.

[0110] FIGS. **10** to **17** are views illustrating a process in which the locking member locks the cover in the display apparatus according to one or more embodiments. FIGS. **10**, **12**, **14**, and **16** are perspective views, and FIGS. **11**, **13**, **15**, and **17** are side views. FIG. **18** is a side cross-sectional view of the display apparatus according to one or more embodiments.

[0111] Referring to FIGS. **10** and **11**, in the display apparatus according to one or more embodiments, the cap cover **600** may be moved toward the side wall **311** to be coupled to the power connector **360**.

[0112] FIGS. **12** and **13** illustrate that the cap cover **600** is coupled to the power connector **360**. At this time, the cap cover **600** may be positioned in a first position, which is a position before being rotated. Once the cap cover **600** is coupled to the power connector **360**, the cap cover **600** may be rotated. For example, the cap cover **600** may be rotated in a first direction to be primarily locked to the power connector **360**.

[0113] Referring to FIGS. **14** and **15**, the cap cover **600** may be in a state in which the mounting protrusion **630** and the mounting recess **363** are coupled by rotation of the cap cover **600**. The rotated position of the cap cover **600** may be a second position. In the second position, the cap cover **600** may be primarily locked. For example, the user may primarily lock the cap cover **600** to the power connector **360** by grasping the handle **640** and/or the cover portion **610** of the cap cover **600** and rotating the cap cover **600**. After the cap cover **600** is primarily locked to the power connector **360**, the locking member **500** may slide downward to secondarily lock the cap cover **600** to prevent the primary locking of the cap cover **600** from being released. For example, the locking member **500** may slide toward the cap cover **600** to secondarily lock the cap cover **600** to prevent the cap cover **600** from rotating in a second direction opposite to the first direction. For example, the user may slide the locking member **500** by pushing the handle **550** of the locking member **500** downward. Alternatively, the user may move the locking member **500** by grasping the reinforcing portion **542**. At this point, the screw **560** may be slightly loosened from the second fixing member **370**.

[0114] Referring to FIGS. **16** to **18**, the cap cover **600** may be secondarily locked as the locking member **500** is moved toward the cap cover **600**. The contact portion **541** may contact the retaining protrusion **620** protruding from the cap cover **600** to prevent the locking member **500** from rotating in the second direction. At this point, the screw **560** may be tightened and fully engaged with the second fixing member **370**. The screw **560** may be tightened or loosened using a tool, such as a screwdriver, rather than the user's hand.

[0115] In the display apparatus according to one or more embodiments, the cap cover **600** may be coupled to the power connector **360** coupled to the side wall **311**, and the locking member **500** may press the cap cover **600** at the perimeter of the cap cover **600** to prevent the cap cover **600** from rotating.

[0116] To remove the cap cover **600**, the process described above may be performed in reverse order. For example, the screw **560** may be slightly loosened by using a screwdriver, the locking member **500** may be slid upward, and then the cap cover **600** may be rotated in the second

direction, thereby releasing the cap cover **600** from the power connector **360**.

[0117] Meanwhile, the second fixing member **370** may be inserted through the side wall **311** of the circuit case **300**. The second fixing member **370** may be inserted into the fixing hole **311b** formed in the one wall **311**.

[0118] The screw **560** may be inserted into the fixing portion **371** of the second fixing member **370**.

[0119] The second fixing member **370** may further include a supporting portion **372**. The supporting portion **372** may support the locking member **500** when the locking member **500** is coupled to the side wall **311** via the screw **560**. The supporting portion **372** may prevent the locking member **500** from directly contacting the circuit case **300**. The supporting portion **372** may be disposed between the side wall **311** of the circuit case **300** and the locking member **500**.

[0120] As described above, according to one or more embodiments of the present disclosure, the cap cover **600** connected to the power connector **360** may be securely locked. As a result, the user cannot manually remove the cap cover **600**, thereby satisfying the safety standards. Nevertheless, the simple structure described above may allow for easy removal, providing the advantage of improving workability.

[0121] In addition, when power-related repairs within the display module **100** are needed, the cap cover **600** connected to the power connector **360** may be simply removed using only a tool such as a screwdriver.

[0122] FIGS. **19** to **22** are views illustrating a process in which the locking member locks the cover in the display apparatus according to one or more embodiments.

[0123] Referring to FIGS. **19** to **22**, the display apparatus according to one or more embodiments may further include the wire conduit **700**. The wire conduit **700** may be connected to the circuit case **300** to supply power to the display module **100**. For example, the wire conduit **700** may be coupled to the power connector **360** arranged on the side wall **311** of the circuit case **300**.

[0124] The wire conduit **700** may include a wire conduit connecting portion **711**. The wire conduit connecting portion **711** may be arranged to correspond to the connecting portion **361** of the power connector **360**. The wire conduit connecting portion **711** may include a connecting protrusion **711a**. The connecting protrusion **711a** may be inserted into the connecting groove **361a** arranged in the connecting portion of the power connector **360** (see FIG. **18**). The connecting protrusion **711a** and the connecting groove **361a** may be formed corresponding to each other. Conversely, it is also possible that a connecting groove is formed in the wire conduit connecting portion **711**, and a connecting protrusion is formed on the power connector **360**.

[0125] The wire conduit **700** may further include a wire conduit cover **712**. The wire conduit cover **712** may cover the wire conduit connecting portion **711**. For example, the wire conduit cover **712** may cover a perimeter of the wire conduit connecting portion **711**. The wire conduit cover **712** may be formed corresponding to a perimeter of the wire conduit connecting portion **711**. The wire conduit cover **712** may include an inner surface **712b** and an outer surface **712a**.

[0126] The wire conduit cover **712** may include a mounting protrusion **730**. The mounting protrusion **730** may protrude from the inner surface **712b** of the wire conduit cover **712** and be mounted to the power connector **360**. When the wire conduit cover **712** is primarily locked to the power connector **360**, the mounting protrusion **730** may be mounted in the mounting recess **363**. The mounting protrusion **730** may be provided in a plurality.

[0127] The wire conduit cover **712** may include the retaining protrusion **720**. The retaining protrusion **720** may protrude from the outer surface **712a** of the wire conduit cover **712** and be retained and secondarily locked by the locking member **500**. After the wire conduit cover **712** is primarily locked to the power connector **360**, the wire conduit cover **712** may be secondarily locked as the locking member **500** slides to contact the mounting protrusion **730** and the locking portion **530**. The retaining protrusions **720** may be provided in a plurality. The retaining protrusion **720** may protrude from the outer surface **712a** of the wire conduit cover **712** on both sides.

[0128] Referring to FIG. **19**, in the display apparatus according to one or more embodiments, the

wire conduit **700** may be moved toward the side wall **311** to be coupled to the power connector **360**.

[0129] Referring to FIG. **20**, the wire conduit **700** may be coupled to the power connector **360**. At this time, the wire conduit cover **712** may be positioned in a first position, which is a position before being rotated. Once the wire conduit cover **712** is coupled to the power connector **360**, the wire conduit cover **712** may be rotated. For example, the wire conduit cover **712** may be rotated in a first direction to be primarily locked to the power connector **360**.

[0130] Referring to FIG. **21**, the wire conduit cover **712** may be in a state in which the mounting protrusion **730** and the mounting recess **363** are coupled by rotation of the wire conduit cover **712**. The rotated position of the wire conduit cover **712** may be a second position. In the second position, the wire conduit cover **712** may be primarily locked. For example, the user may primarily lock the wire conduit cover **712** to the power connector **360** by grasping the wire conduit cover **712** and rotating the wire conduit cover **712**. After the wire conduit cover **712** is primarily locked to the power connector **360**, the locking member **500** may slide downward to secondarily lock the wire conduit cover **712** to prevent the primary locking of the wire conduit cover **712** from being released. For example, the locking member **500** may slide toward the wire conduit cover **712** to secondarily lock the wire conduit cover **712** to prevent the wire conduit cover **712** from rotating in a second direction opposite to the first direction. For example, the user may slide the locking member **500** by pushing the handle **550** of the locking member **500** downward. Alternatively, the user may move the locking member **500** by grasping the reinforcing portion **542**. At this point, the screw **560** may be slightly loosened from the second fixing member **370**.

[0131] Referring to FIG. **22**, the wire conduit cover **712** may be secondarily locked as the locking member **500** is moved toward the wire conduit cover **712**. The contact portion **541** may contact the retaining protrusion **720** protruding from the wire conduit cover **712** to prevent the locking member **500** from rotating in the second direction. At this point, the screw **560** may be tightened and completely engaged with the second fixing member **370**. The screw **560** may be tightened or loosened using a tool such, as a screwdriver, rather than the user's hand.

[0132] In the display apparatus according to one or more embodiments, the wire conduit **700** may be coupled to the power connector **360** coupled to the side wall **311**, and the locking member **500** may press the wire conduit cover **712** at the perimeter of the wire conduit cover **712** to prevent the wire conduit cover **712** from rotating.

[0133] To remove the wire conduit cover **712**, the process described above may be performed in reverse order. For example, the screw **560** may be slightly loosened by using a screwdriver, the locking member **500** may be slid upward, and then the wire conduit cover **712** may be rotated in the second direction, thereby releasing the wire conduit **700** from the power connector **360**.

[0134] While various embodiments have been described above with reference to the drawings, the present disclosure is not limited thereto, and any combination or substitution of components as appropriate is included within the scope of the present disclosure. In some embodiments, modifications such as combinations, changes in the order of processes, and various changes in design may be made on the basis of knowledge of a person skilled in the art, and such modified embodiments are within the scope of the present disclosure and the appended claims.

Claims

1. A display apparatus comprising: a display module; a cabinet to which the display module is coupled thereto; a circuit case coupled to the cabinet and configured to receive a printed circuit board; a power connector on one wall of the circuit case and configured to supply power to the printed circuit board; a cover detachably coupled to the power connector; and a locking member coupled to the one wall of the circuit case and configured to lock the cover to prevent the cover from being removed by preventing rotation of the cover in a state in which the cover is coupled to

the power connector.

2. The display apparatus of claim 1, wherein the locking member is configured to slide toward the cover to contact the cover.

3. The display apparatus of claim 2, wherein the locking member comprises: a fastening portion fastened to the one wall of the circuit case; and a locking portion connected to the fastening portion, contactable with the cover, and configured to prevent the rotation of the cover.

4. The display apparatus of claim 3, wherein the cover comprises: a cover portion having a shape corresponding to a shape of the power connector and configured to receive the power connector in a state in which the cover is coupled to the power connector, and a retaining protrusion protruding from an outer surface of the cover portion, contactable with the locking portion, and configured to prevent the rotation of the cover.

5. The display apparatus of claim 4, wherein the cover portion comprises: a first cover portion aligned with the one wall in a state in which the cover is coupled to the power connector; and a second cover portion extending from the first cover portion and covering a perimeter of the power connector, and wherein the retaining protrusion protrudes from an outer surface of the second cover portion.

6. The display apparatus of claim 5, further comprising a screw configured to be inserted into the one wall through the fastening portion and fasten the locking member to the one wall of the circuit case, wherein the locking member is slidable relative to the screw and configured to lock the cover in a state in which the locking member is slid relative to the screw.

7. The display apparatus of claim 6, wherein the locking member further comprises an inclined portion inclined from the fastening portion toward the locking portion in a direction away from the one wall.

8. The display apparatus of claim 7, wherein the cover is configured to be rotated in a first direction to lock the power connector when the cover is coupled to the power connector, and wherein the locking member is configured to slide toward the cover and contact the cover to lock the cover and prevent the cover from rotating in a second direction opposite to the first direction.

9. The display apparatus of claim 8, wherein the cover further comprises a mounting protrusion protruding from an inner surface of the second cover portion.

10. The display apparatus of claim 9, wherein the power connector comprises: a connecting portion; an outer wall covering a perimeter of the connecting portion; and a mounting recess on the outer wall such that the mounting protrusion is mountable in a state in which the cover is primarily locked to the power connector.

11. The display apparatus of claim 1, further comprising a wire conduit connectable to the power connector and configured to supply power to the display module from an outside, wherein the wire conduit comprises: a wire conduit connecting portion connected to the power connector; and a wire conduit cover configured to cover a perimeter of the wire conduit connecting portion, and wherein, in a state in which the cover is removed from the power connector, the wire conduit is coupled to the power connector and the locking member contacts the wire conduit cover and is configured to prevent rotation of the wire conduit cover.

12. The display apparatus of claim 11, wherein the locking member comprises: a fastening portion fastened to the one wall; an inclined portion inclined from the fastening portion in a direction away from the one wall; and a locking portion connected to the inclined portion on a side far from the one wall and contactable with the wire conduit cover and configured to prevent the rotation of the wire conduit cover.

13. The display apparatus of claim 12, wherein the wire conduit cover comprises a retaining protrusion protruding from an outer surface of the wire conduit cover and contactable with the locking portion and configured to prevent the rotation of the wire conduit cover.

14. The display apparatus of claim 13, wherein the wire conduit cover further comprises a fastening protrusion protruding from an inner surface of the wire conduit cover such that the wire conduit

cover is locked to the power connector.

15. The display apparatus of claim 14, wherein in a state in which the wire conduit and the power connector are connected to each other, the wire conduit cover is configured to be rotated in a first direction and locked to the power connector, and wherein the locking member is configured to slide toward the wire conduit cover to lock the wire conduit cover and prevent the wire conduit cover from rotating in a second direction opposite to the first direction.
